

# The birth of the Universe

## The Big Bang

By the mid-20th century, there were two competing theories for the origin of the Universe. The Steady State theory held that matter is continuously created as the Universe expands, the overall density of the Universe remains the same, and the Universe has existed forever.

Most scientists now believe that we live in a finite expanding Universe which has not existed forever, and that all the matter, energy and space in the Universe was once squeezed into an infinitesimally small volume, which erupted in a cataclysmic “explosion” which has become known as the Big Bang. The term, ‘The Big Bang’, was invented by the English astronomer Fred Hoyle during a 1949 radio broadcast as a derisive description of a theory he disagreed with.

The theory implies that space, time, energy and matter all came into being at an infinitely dense, infinitely hot gravitational singularity, and began expanding everywhere at once. Modern measurements place this event at approximately 13.8 billion years ago and thus this is considered the age of the Universe. The model offers a comprehensive explanation for a broad range of observed phenomena, including the abundance of light elements, the cosmic microwave background, large scale structure, and Hubble’s Law.

According to prevailing scientific thinking, if we were to look at the Universe one second after the Big Bang, what we would see is a 10-billion degree sea of neutrons, protons, electrons, anti-electrons (positrons), photons, and neutrinos. Some time later we would see the Universe cool; the neutrons would decay forming protons and electrons or would combine with protons to make deuterium (an isotope of hydrogen). Continued cooling would eventually lead to electrons combining with nuclei to form simple atoms. Giant clouds of these primordial elements would later coalesce through gravity to form matter and eventually stars and galaxies.

## The Cosmic Microwave Background

In 1963, Arno Penzias and Robert Wilson were studying faint microwave signals from the Milky Way galaxy. They found a mysterious noise of unknown origin. At first the noise was thought to be interference caused by pigeon droppings on the antenna equipment. Pigeons were trapped and dung was cleaned from the antenna. Ultimately Penzias and Wilson realized that the noise was an actual signal. Penzias and Wilson theorised that if the Big Bang theory was correct, the Universe would be filled with background radiation left over from the creation event. This radiation is now called the cosmic microwave background, or CMB.

The CMB was created at a time in cosmic history called the Recombination Era. About 378,000 years after the Big Bang the Universe was a hot, dense and opaque plasma containing both matter and energy. Photons could not travel freely, so no light escaped. The Universe then cooled to a temperature of about 2,700 °C which enabled electrons and protons to form stable hydrogen atoms. This process released photons, creating the radiation that is now called the CMB.

### The Expanding Universe

In 1925, the American astronomer Edwin Hubble stunned the scientific community by demonstrating that there was more to the Universe than just our Milky Way galaxy and that there were in fact many separate islands of stars - thousands, perhaps millions of them, and many of them huge distances away from our own. Then, in 1929, Hubble announced a further dramatic discovery. Using improved telescopes, Hubble observed that the light coming from these galaxies was shifted a little toward the red end of the spectrum, which indicated that the galaxies were moving away from us. When the source of a wave is moving toward an observer the wavelength is shortened (called 'blue shift') and when the source is moving away from the observer the wavelength is lengthened ('red shift'). Hubble observed that the emission spectra of distant galaxies were all red-shifted, and the further away they were the greater the red shift.

After a detailed analysis Hubble concluded that the galaxies and clusters of galaxies were in fact flying apart from each other at a speed that was in direct proportion to its distance. This is known as Hubble's Law. A galaxy that is twice as far away as another is receding twice as fast, one ten times as far away if receding ten times as fast, etc. The law is usually stated as  $v = H_0 D$ , where  $v$  is the velocity of recession,  $D$  is the distance of the galaxy from the observer and  $H_0$  is the Hubble constant which is around 22 km/s/million light years.

Individual galaxies themselves are not expanding but space itself is expanding. Imagine a balloon inflating, if tiny dots are painted on the balloon to represent galaxies, then as the balloon expands so the distance between the dots increases and the further apart the dots the faster they move apart. In such an expansion the Universe continues to look more or less the same from every galaxy.



Edwin Hubble

### Comprehension task

Draw a time line starting at the Big Bang to show when and how matter evolved from sub-atomic particles to heavy nuclei.

At points along the time line, there are great changes in the nature of the Universe and the matter in it. Explain what caused these changes.

### The standard model explained

#### Notes

#### Main points

- The Big Bang theory describes the early development of the Universe, including the formation of subatomic particles from energy and the subsequent formation of atomic nuclei.
- A variety of evidence supports the Big Bang theory, including cosmic background radiation, the abundance of light elements and the red shift of light from galaxies that obey Hubble's law.
- Alternative theories exist, including the Steady State theory, but the Big Bang theory is the most widely accepted theory today.
- The standard model is used to describe the evolution of forces and the creation of matter in the Big Bang theory.
- The expansion of the Universe can be explained by Hubble's law and cosmological concepts, such as red shift and the Big Bang theory.

The big questions are, "What is the world made of?" and "What holds it together?"

- The standard model is the most complete explanation of the fundamental particles and interactions to date.
- The standard model is based on the premise that all matter in the Universe is made up from elementary matter particles called quarks and leptons; quarks experience the strong nuclear force, leptons do not.
- The world is made of six quarks and six leptons. Everything we see is a conglomeration of quarks and leptons.
- Baryons are composite particles made up of quarks.
- Names and descriptions are only a small part of any physical theory; the concepts, rather than physics vocabulary, are the critical elements.
- A particle's state (set of quantum numbers) can affect how it interacts with other particles.
- Lepton number and baryon number are conserved in all reactions between particles; these conservation laws can be used to support or invalidate proposed reactions.
- There are four fundamental forces, and force carrier particles are associated with each force.
- The standard model explains three of the four fundamental forces (strong, weak and electromagnetic forces) in terms of an exchange of force carrying particles called gauge bosons; each force is mediated by a different type of gauge boson.
- High-energy particle accelerators are used to test theories of particle physics, including the standard model.
- The magnitude of the force experienced by a particle travelling in a magnetic field depends on the charge of the particle, the velocity of the particle and the strength of the field.

Physicists have developed a theory called the standard model that explains what the world is made of, and what holds it together. Developed since the middle of the 20th century, the theory holds that all matter in the Universe is made up of combinations of elementary particles called quarks and leptons. It describes all of the particles and the interactions between. It explains the strong, weak and electromagnetic forces as a result of the exchange of various types of force carrier particles between the elementary matter particles. All the known matter particles are composites of quarks and leptons, and they interact by exchanging force carrier particles. The Standard Model has been verified by experiments; all the particles predicted by this theory have been found. But it does not explain everything. For example, gravity is not included in the Standard Model. Dark energy and dark matter are not yet understood, but are believed to be cosmological properties required to explain the expanding Universe.

### Structure within the atom

Protons and neutrons vibrate within the nucleus, and quarks vibrate within the protons and neutrons.

Figure 15.1 is quite distorted. If we drew the atom to scale and made protons and neutrons a centimetre in diameter, then the electrons and quarks would be smaller than the diameter of human hair and the entire atom's diameter would be greater than the length of thirty football fields; 99.999999999999% of an atom's volume is just empty space!

While an atom is tiny, the nucleus is ten thousand times smaller than the atom and the quarks and electrons are at least ten thousand times smaller than that. We don't know exactly how small quarks and electrons are; they are definitely smaller than  $10^{-18}$  metres, or they might literally be points, but we just do not know. It is also possible that quarks and electrons are not fundamental after all, and will turn out to be made up of other, more fundamental articles ... and on it goes. Relative sizes are shown in Figure 15.2.

Physicists are constantly look for new particles. When they find a new particle, they find out its properties, trying to find patterns that tell us what the fundamental building blocks of the Universe are, and how they interact. We now know of about two hundred particles most of which aren't fundamental. These particles are named using letters from the Greek and Roman alphabets. Of course, the names of particles are but a small part of any physical theory. You should not be discouraged if you have trouble remembering them. Take heart: even the great Enrico Fermi once said to his student (and future Nobel Laureate) Leon Lederman, Young man, if I could remember the names of these articles, I would have been a botanist!"

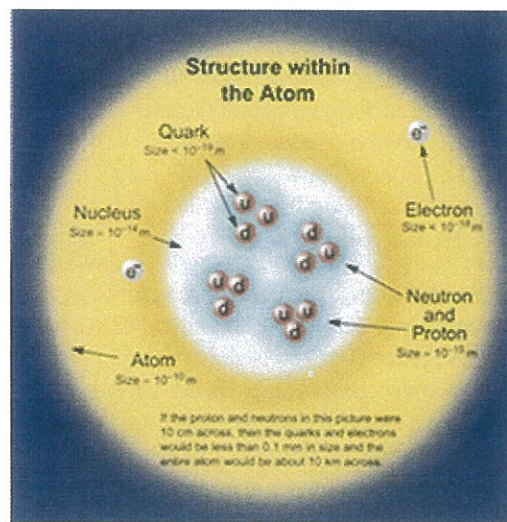


Figure 15.1: This is the modern atom model. Electrons are in constant motion around the nucleus, protons and neutrons vibrate within the nucleus, and quarks vibrate within the protons and neutrons (ParticleAdventure.org).

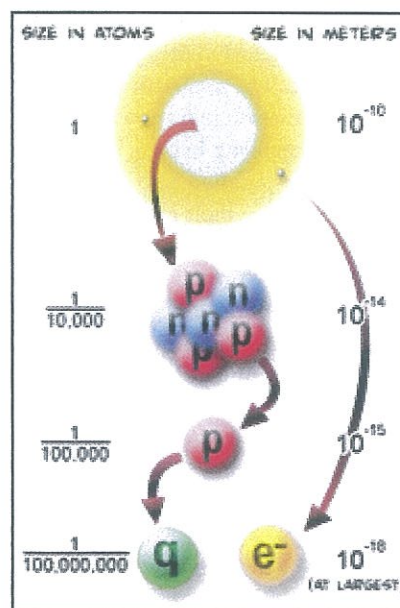


Figure 15.2

### Notes

# The standard model explained

## Notes

### The Pauli Exclusion Principle

We can use these quantum particle properties to categorise the particles we find. Physicists once thought that no two particles in the same quantum state could exist in the same place at the same time. This is called the Pauli Exclusion Principle, and it explains much about chemistry. But certain particles do not obey this principle. Particles that do obey the Pauli Exclusion Principle are called fermions, and those that do not are called bosons.

Imagine a large family of identical fermion siblings spending the night at the Fermion Motel, and another large family of identical boson siblings spending the night at the Boson Inn. Fermions behave like squabbling siblings, and not only refuse to share a room but also insist on rooms as far as possible from each other. On the other hand, boson siblings prefer to share the same room.

### Fermions and bosons

These are the matter particles consisting of six quarks and the six leptons. They are grouped together under this name because they all obey the Pauli Exclusion Principle and all have an anti-particle.

A fermion is any particle whose spin is an odd half-integer (such as  $1/2$ ,  $3/2$ , and so forth). Quarks and leptons, as well as most composite particles, such as protons and neutrons, are fermions. For reasons we do not fully understand, having odd half-integer spin means that fermions obey the Pauli Exclusion Principle (i.e. two fermions cannot co-exist in the same state at same location at the same time).

Bosons have an integer spin (0, 1, 2...). All force carrier particles are bosons, as are those composite particles with an even number of fermion particles (such as mesons).

### Matter and antimatter

For every type of matter particle we know, there exists a corresponding antimatter particle, or antiparticle. The antiparticle is identical in mass to the particle from which it derives its name but opposite in sign. The antiparticle of the electron is called a positron. Antiparticles behave just like their corresponding matter particles, except they have opposite charges. For instance, a proton is electrically positive whereas an antiproton is electrically negative. When a matter particle and antimatter particle meet, they annihilate into pure energy.

The Universe appears to be composed entirely of matter. If antimatter and matter are exactly equal but opposite, then why is there so much more matter in the Universe than antimatter? Well ... that is a question that keeps physicists up at night.

The usual symbol for an antiparticle is a bar over the corresponding particle symbol. For example, the "up quark"  $u$  has an "up" antiquark" designated by  $\bar{u}$ , pronounced u-bar. The antiparticle of a quark is an antiquark, the antiparticle of a proton is an antiproton, and so on. The antielectron is called a positron and is designated  $e^+$ .

|         |                              |                            |                            |                     |                         |
|---------|------------------------------|----------------------------|----------------------------|---------------------|-------------------------|
| charge  | $+\frac{2}{3}$               | $+\frac{2}{3}$             | $+\frac{2}{3}$             | 0                   | 0                       |
|         | <b>u</b><br>up               | <b>c</b><br>charm          | <b>t</b><br>top            | <b>g</b><br>gluon   | <b>H</b><br>Higgs boson |
|         | $-\frac{1}{3}$               | $-\frac{1}{3}$             | $-\frac{1}{3}$             | 0                   |                         |
| QUARKS  | <b>d</b><br>down             | <b>s</b><br>strange        | <b>b</b><br>bottom         | $\gamma$<br>photon  |                         |
|         | -1                           | -1                         | -1                         | 0                   |                         |
|         | <b>e</b><br>electron         | $\mu$<br>muon              | $\tau$<br>tau              | <b>Z</b><br>Z boson |                         |
| LEPTONS | 0                            | 0                          | 0                          | $\pm 1$             |                         |
|         | $\nu_e$<br>electron neutrino | $\nu_\mu$<br>muon neutrino | $\nu_\tau$<br>tau neutrino | <b>W</b><br>W boson |                         |
|         |                              |                            |                            |                     | GAUGE BOSONS            |

Figure 15.3: Quarks, leptons and bosons (ParticleAdventure.org)

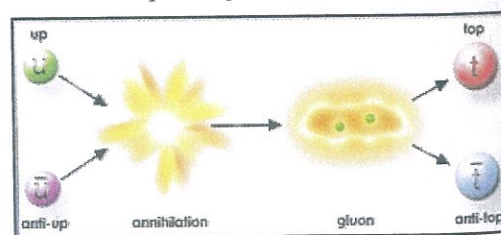


Figure 15.4: Fermions and Bosons (ParticleAdventure.org)

## Quarks

There are six quarks. They are never found alone but as combinations of two or three in particles called hadrons. Three quarks having charge of  $+2/3$  are called up, charm and top; and three having charge  $-1/3$  are called down, strange and bottom. The up and down quarks together account for both the proton and neutron.

All matter from galaxies to mountains to molecules is made from quarks and leptons. But that is not the whole story. Quarks behave differently than leptons, and for each kind of matter particle there is a corresponding antimatter particle. Quarks are one type of matter particle. Most of the matter we see around us is made from protons and neutrons, which are composed of quarks.

There are six quarks, but physicists usually talk about them in terms of three pairs: up/down, charm/strange, and top/bottom. (Also, for each of these quarks, there is a corresponding antiquark.) Quarks have the unusual characteristic of having a fractional electric charge, unlike the proton and electron, which have integer charges of  $+1$  and  $-1$  respectively. Quarks also carry another type of charge called colour charge, which we will discuss later. The most elusive quark, the top quark, was discovered in 1995 after its existence had been theorized for 20 years.

The naming of quarks.

In 1964, Murray Gell-Mann and George Zweig suggested that hundreds of the particles known at the time could be explained as combinations of just three fundamental particles. Gell-Mann chose the name “quarks,” pronounced “kworks,” for these three particles, a nonsense word used by James Joyce in the novel *Finnegan’s Wake*. In order to make their calculations work, the quarks had to be assigned fractional electrical charges of  $2/3$  and minus  $1/3$ . Quarks are never observed by themselves, and so initially these quarks were regarded as fiction. Experiments have since convinced physicists that not only do quarks exist, but there are six of them, not three.

There are six ‘flavors’ of quarks which just means different kinds. The two lightest are called up and down. The third quark is called strange. It was named after the “strangely” long lifetime of the K particle, the first composite particle found to contain this quark. The fourth quark type, the charm quark was discovered in 1974 almost simultaneously at both the Stanford Linear Accelerator Centre (SLAC) and at Brookhaven National Laboratory.

The fifth and sixth quarks ‘top’ and ‘bottom’ were sometimes called ‘truth’ and ‘beauty’. The bottom quark was first discovered at Fermi National Lab (Fermilab) in 1977. The top quark was discovered last, also at Fermilab, in 1995. It is the most massive quark. It had been predicted for a long time but had never been observed successfully until then.

## Hadrons

These are particles made up of two or more quarks, are capable of existence on their own (as opposed to in combination) and can interact strongly with each other. Quarks only exist in groups with other quarks and are never found alone. Composite particles made of quarks are called **Hadrons**. Only a very small part of the mass of a hadron is due to the quarks in it.

Although individual quarks have fractional electrical charges, they combine such that hadrons have a net integer electric charge. Another property of hadrons is that they have no net colour charge even though the quarks themselves carry colour charge.

There are two classes of hadrons: baryons and mesons:

- A baryon is any hadron made of three quarks (qqq). Protons are baryons, made of two up quarks and one down quark (uud). So are neutrons (udd).
- A meson is a hadron made from a quark and its anti-quark. One example of a meson is a pion (+), which is made of an up quark and a down antiquark. The antiparticle of a meson just has the quark and antiquark switched, so an antipion (-) is made of a down quark and an up antiquark. Mesons, made of a particle and an antiparticle, are very unstable.

## Notes

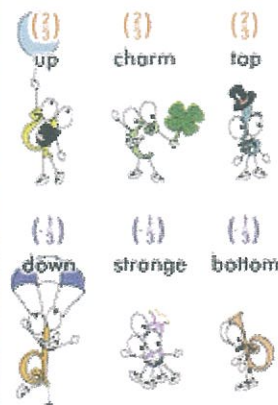


Figure 15.5: Six quarks  
(ParticleAdventure.org)

# The standard model explained

## Notes

The K meson lives much longer than most mesons, which is why it was called “strange” and gave this name to the strange quark, one of its components.

### Leptons

The other group of matter particles is the leptons. There are six leptons, three being electrically charged and three being neutral. Leptons appear to be point-like particles without internal structure. The best known lepton is the electron ( $e^-$ ). The other two charged leptons are the muon and the tau, which like electrons are charged but with a lot more mass. The other leptons are the three types of neutrinos. These have no electrical charge, very little mass, and are very hard to detect.

Each pair of leptons is intimately connected to a pair of quarks – the electron and its neutrino are connected to the up and down quarks, the muon and its neutrino to the strange and charm quarks and the tau and its neutrino to the top and bottom quarks.

Leptons are solitary particles. Each lepton has a corresponding antilepton. The heavier leptons, the muon and the tau, are not found in ordinary matter at all. This is because when they are produced they very quickly decay, or transform, into lighter leptons. Sometimes the tau lepton will decay into a quark, an antiquark, and a tau neutrino. Electrons and the three kinds of neutrinos are stable.

When a heavy lepton decays, one of the particles it decays into is always its corresponding neutrino. The other particles could be a quark and its antiquark, or another lepton and its antineutrino.

Physicists have observed that some types of lepton decays are possible and some are not. In order to explain this, they divided the leptons into three lepton families: the electron and its neutrino, the muon and its neutrino, and the tau and its neutrino. The number of members in each family must remain constant in a decay; a particle and an antiparticle in the same family “cancel out” to make the total of them equal zero. Although leptons are solitary, they are always loyal to their families!

### Lepton type conservation

Leptons are divided into three lepton families: the electron and its neutrino, the muon and its neutrino, and the tau and its neutrino. The terms “electron number,” “muon number,” and “tau number” to refer to the lepton family of a particle. Electrons and their neutrinos have electron number +1, positrons and their antineutrinos have electron number -1, and all other particles have electron number 0. Muon number and tau number operate analogously with the other two lepton families.

| equation        | muon  | muon neutrino | electron | e <sup>+</sup> antineutrino |
|-----------------|-------|---------------|----------|-----------------------------|
|                 | $\mu$ | $\nu_\mu$     | $e^-$    | $\bar{\nu}_e$               |
| electron number | 0     | 0             | 1        | -1                          |
| muon number     | 1     | 1             | 0        | 0                           |
| Tau number      | 0     | 0             | 0        | 0                           |

One important thing about leptons, then, is that electron number, muon number, and tau number are always conserved when a massive lepton decays into smaller ones. Let's take an example decay. A muon decays into a muon neutrino, an electron, and an electron antineutrino (*see left*)

As you can see, electron, muon, and tau numbers are conserved. These and other conservation laws are what we believe define whether or not a given hypothetical lepton decay is possible.

### Neutrinos

Neutrinos are a type of lepton. Since they have no electrical or strong charge they almost never interact with any other particles. Most neutrinos pass right through the earth without ever interacting with a single atom of it.

They are produced in a variety of interactions, especially in particle decays. In fact, it was through a careful study of radioactive decays that physicists hypothesized the neutrino's existence. For example:

- In a radioactive nucleus, a neutron at rest (zero momentum) decays, releasing a proton and an electron.
- Because of the law of conservation of momentum, the resulting products of the decay must have a total momentum of zero, which the observed proton and electron clearly do not. (Furthermore, if there are only two decay products, they must come out back-to-back.)
- Therefore, we need to infer the presence of another particle with appropriate momentum to balance the event.
- We hypothesize that an antineutrino was released; experiments have confirmed that this is indeed what happens.

Because neutrinos were produced in great abundance in the early Universe and rarely interact with matter there are a lot of them in the Universe. Their tiny mass but huge numbers may contribute to the total mass of the Universe and affect its expansion.

### Gauge bosons

Gauge bosons are massless and are referred to as the force carriers because they are the particles responsible for mediating the strong and weak interactions. Particles subject to either the strong or the weak interaction exchange bosons as they interact. The  $W^+$ ,  $W^-$  and  $Z^0$  mediate the weak interactions between the different flavours of quarks and leptons, the eight gluons mediate the strong force between the quarks and the photon mediates the electromagnetic interaction.

### Photons

Photons are different from the other bosons and particles because they do not have a rest mass and travel at the speed of light. Depending on its energy a photon can be transformed into a particle/anti-particle pair. Conversely, a collision between a particle and its anti-particle will result in the annihilation of those particles and the formation of photons. The relationship between the mass and energy interchanged can be calculated from Einstein's equation  $E = mc^2$ , where all of the energy becomes mass or all of the mass will become energy.

### Flavour

An alternative name is generation. Flavour is a property that separates the different types of leptons and quarks. Each of the charged leptons and its associated neutrino has a distinct flavour, as do the three flavours of quarks, with two per flavour for both. The first flavour of both leptons and quarks (the electrons, protons and neutrons) does not undergo decay, with the electrons orbiting the atomic nuclei containing the protons and neutrons which are composed of up and down quarks. The second flavour (charm and strange quark; muon and muon neutrino) and third flavour (top and bottom quark; tau and tau neutrino) have short half-lives and are only observed in very high energy particle accelerators. The masses of the particles in each successive flavour are greater than that of corresponding particles in the flavour before. This is why successive flavours were discovered only when more powerful particle accelerators were built.

### Notes

# The standard model explained

## Notes

### The four interactions

Matter is made of quarks and leptons but what holds all these particles together?

Matter exists because of the ways in which the fundamental particles interact. These interactions include attractive and repulsive forces, decay, and annihilation. There are four fundamental interactions between particles, and all forces in the world can be attributed to these four interactions. Any force you can think of - friction, magnetism, gravity, and so on - is caused by one of these four fundamental interactions.

How do things interact without touching! How do two magnets “feel” each other’s presence and attract or repel accordingly? How does the Sun attract the Earth? Generally the answers given to these questions are “magnetism” and “gravity,” but how do these interactions work? At a fundamental level, a force is not just something that happens to particles, it is a thing that is passed from one particle to another.

Interactions that affect matter particles arise from an exchange of force carrier particles, a different type of particle altogether. What we normally think of as “forces” are actually the effects of force carrier particles on matter particles. We see examples of attractive forces in everyday life (such as magnets and gravity), and so we generally take it for granted that an object’s presence can just affect another object. The deeper question, “How can two objects affect one another without touching?” leads us to propose that the invisible force could be an exchange of force carrier particles. Particle physicists can explain the force of one particle acting on another to incredible precision through the exchange of force carrier particles.

One important thing to know about force carriers is that a particular force carrier particle can only be absorbed or produced by a matter particle which is affected by that particular force. For instance, electrons and protons have electric charge, so they can produce and absorb the electromagnetic force carrier, the photon. Neutrinos, on the other hand, have no electric charge, so they cannot absorb or produce photons.

### The electromagnetic force

The electromagnetic force causes like-charged things to repel and oppositely-charged things to attract. Many everyday forces, such as friction, and even magnetism, are caused by the electromagnetic force. For instance, the force that keeps you from falling through the floor is the electromagnetic force which causes the atoms making up the matter in your feet and the floor to resist being displaced.

The carrier particle of the electromagnetic force is the photon. Photons of different energies span the electromagnetic spectrum of x rays, visible light, radio waves, and so forth. Photons have zero mass, as far as we know, and always travel at the “speed of light”,  $c$ , which is about 300,000,000 metres per second in a vacuum.

Atoms usually have the same numbers of protons and electrons. They are electrically neutral, therefore, because the positive protons cancel out the negative electrons. Since they are neutral, what causes them to stick together to form stable molecules? The answer is that the charged parts of one atom can interact with the charged parts of another atom. This allows different atoms to bind together, an effect called the residual electromagnetic force. This is what allows atoms to bond and form molecules, allowing the world to stay together and create the matter you interact with all of the time. All the structures of the world exist simply because protons and electrons have opposite charges!

### The Strong Nuclear Force

What then binds the nucleus together? The nucleus of an atom consists of a bunch of protons and neutrons crammed together. Since neutrons have no charge and the positively-charged protons repel one another, why doesn’t the nucleus blow apart?

We cannot account for the nucleus staying together with just electromagnetic force. What else could there be?

To understand what is happening inside the nucleus, we need to understand more about the quarks that make up the protons and neutrons in the nucleus. Quarks have electromagnetic charge, and they also have an altogether different kind of charge called colour charge. The force between colour-charged particles is very strong, so this force is called 'the strong force'. The strong force holds quarks together to form hadrons, so its carrier particles are called gluons because they so tightly "glue" quarks together.

### Colour

'Colour' or 'colour charge' is a label and has nothing to do with actual colour. It is a name given to the three states of quarks labelled red, blue and green. Anti-quarks are labelled anti-red, anti-blue and anti-green. Gluons, which hold quarks together, are also coloured. When a gluon is emitted or absorbed by a quark the quark may change colour (state) but the large composite particle is always colourless. In order to be colourless, baryons (protons and neutrons), are comprised of three quarks, and must therefore have one of each colour. Mesons are comprised of two quarks, and must comprise a quark/anti-quark pair.

Quarks and gluons are colour-charged particles, they exchange gluons in strong interactions. When two quarks are close to one another, they exchange gluons and create a very strong colour force field that binds the quarks together. The force field gets stronger as the quarks get further apart. Quarks constantly change their colour charges as they exchange gluons with other quarks.

There are three colour charges and three corresponding anti-colour (complementary colour) charges. Each quark has one of the three colour charges and each antiquark has one of the three anti-colour charges. Just as a mix of red, green, and blue light yields white light, in a baryon a combination of "red," "green," and "blue" colour charges is colour neutral, and in an antibaryon "anti-red," "anti-green," and "anti-blue" is also colour neutral. Mesons are colour neutral because they carry combinations such as "red" and "anti-red."

Because gluon-emission and -absorption always changes colour, and -in addition - colour is a conserved quantity - gluons can be thought of as carrying a colour and an anti-colour charge. Since there are nine possible colour-anti-colour combinations we might expect nine different gluon charges, but it works out that there are only eight combinations.

Colour-charged particles cannot be found individually. For this reason, the colour-charged quarks are confined in groups (hadrons) with other quarks. These composites are colour neutral.

The development of the Standard Model's theory of the strong interactions reflected evidence that quarks combine only into baryons (three quark objects), and mesons (quark-antiquark objects), but not, for example, four-quark objects. Now we understand that only baryons (three different colours) and mesons (colour and anti-colour) are colour-neutral. Particles such as uuu or dddd that cannot be combined into colour-neutral states are never observed.

### Notes

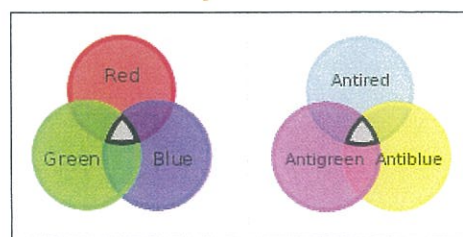


Figure 15.7: Three colours or anti-colours combine to be colourless.

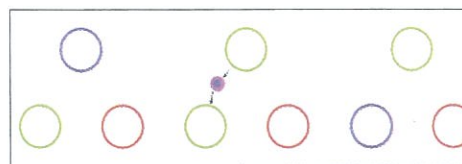


Figure 15.8: A. A hadron with three quarks (red, green, blue) before a colour change

B. Blue quark emits a blue-antigreen gluon

C. Green quark has absorbed the blue-antigreen gluon and is now blue; colour remains conserved

# The standard model explained

## Notes

### Residual strong force

The strong force binds quarks together because quarks have colour charge. But that still does not explain what holds the nucleus together, since positive protons repel each other with electromagnetic force, and protons and neutrons are colour-neutral. So what holds the nucleus together? The answer is that, in short, they don't call it the strong force for nothing. The strong force between the quarks in one proton and the quarks in another proton is strong enough to overwhelm the repulsive electromagnetic force. This is called the residual strong interaction, and it is what "glues" the nucleus together.

### The Weak Force

Weak interactions are responsible for the decay of massive quarks and leptons into lighter quarks and leptons. When fundamental particles decay we observe the particle vanishing and being replaced by two or more different particles. Although the total of mass and energy is conserved, some of the original particle's mass is converted into kinetic energy, and the resulting particles always have less mass than the original particle that decayed. The only matter around us that is stable is made up of the smallest quarks and leptons, which cannot decay any further.

When a quark or lepton changes type (a muon changing to an electron, for instance) it is said to change flavour. All flavour changes are due to the weak interaction. The carrier particles of the weak interactions are the  $W^+$ ,  $W^-$ , and the  $Z$  particles. The  $W$ 's are electrically charged and the  $Z$  is neutral.

### Gravity

What about gravity? Gravity has been considered one of the fundamental interactions, but the Standard Model cannot satisfactorily explain it. This is one of those major unanswered problems in physics today. In addition, the gravity force carrier particle has not been found. Such a particle may someday be found: the graviton.

Fortunately, the effects of gravity are extremely tiny in most particle physics situations compared to the other three interactions, so theory and experiment can be compared without including gravity in the calculations. Thus, the Standard Model works without explaining gravity.

# Experiment 15.1: Detecting the world - wavelength and resolution

## Background

When using waves to detect the physical world the quality of the image is limited by the wavelength you use. Our eyes are attuned to visible light, which has wavelengths of about 0.5 micrometres which means that we can resolve objects larger than about a micrometre. However, the wavelength of visible light is too long to analyse anything smaller than a cell. To observe things under higher magnification, you must use waves with smaller wavelengths. That's why scientists use scanning electron microscopes when studying sub-microscopic things such as viruses. However, even the best scanning electron microscope can only show a fuzzy picture of an atom.

A good example of the wavelength vs. resolution issue is a swimming pool. If you have a swimming pool with waves which are 1 metre apart (a 1 metre wavelength) and push a stick into the water, the pool's waves just pass around the stick because the large wavelength ensures that the waves are not affected by such a tiny target. What physical effect is responsible for this?

All particles have wave properties. So, when using a particle as a probe, we need to use particles with short wavelengths to get detailed information about small things. As a rough rule of thumb, a particle can only probe down to distances equal to the particle's wavelength. To probe down to smaller scales the probe's wavelength has to be made smaller.

## Lab notes

### Part A

Use a pool or a ripple-tank to investigate the relationship between wavelength and resolution. Set up the tank according to the instructions, and begin generating waves.

Insert a thin upright wire into the water, downstream from the wave source – does the wire affect the wave pattern? Would you know the wire was there if you could only observe the wave pattern downstream? Repeat with thicker wires, rods and tubes etc. and with different wavelengths, to obtain a relationship between wavelength and resolution (the ability to observe a small object).

### Part B

Different sized balls can be used to obtain an image of an object. With an object placed on the ground use balls to surround the object and then remove the object to observe the 'image' created. How does the size of the balls affect the detail of the image?

## Notes

# Investigation 15.2: Detecting the world - particle beams

## Background

Most of the experiments that have given rise to our current conception of particle physics have occurred relatively recently. But the story of how physicists experiment to test and create theories in modern particle physics is one which starts less than a hundred years ago. In 1909 a man named Ernest Rutherford set up an experiment to test the validity of the prevailing theory of the atom. In doing so he established a way, by using particle beams, that for the first time physicists could “look into” tiny particles they couldn’t see with microscopes.

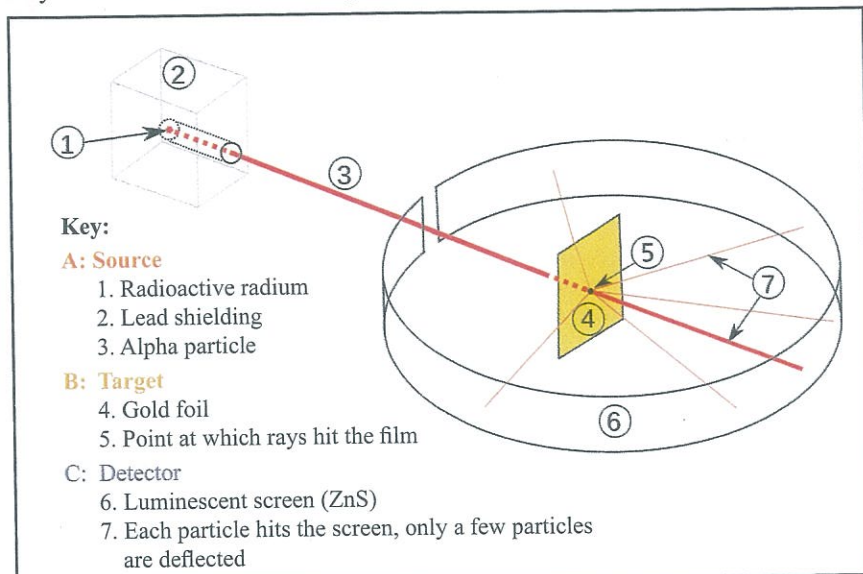


Figure 15.1: Rutherford's experiment

Rutherford's experiment is shown in Figure 15.1. The alpha particles were expected to pass right through the very thin gold foil and make their marks in a small cluster on the screen. If atoms were permeable, neutral balls, then the alpha particles should simply pass through the gold foil and strike the back of the screen. But much to everyone's surprise, some of the alpha particles were deflected at large angles to the foil; some even hit the screen in front of the foil. The deflection of the particles is shown in Figure 15.2.

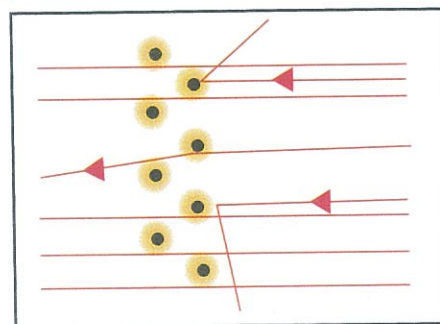
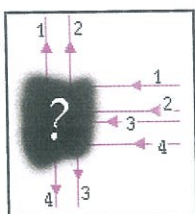


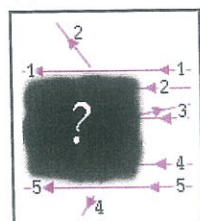
Figure 15.2: Target, gold foil atoms, magnified

## Notes

(a)



(b)



Almost all particle physics experiments today use the same basic elements that Rutherford did:

- A source (in this case, a beam of alpha particles)
- A target (the gold atoms in the foil)
- A detector (the zinc sulfide screen)

In addition, Rutherford established the practise of “seeing” into the sub-atomic realm by using particle beams and particle physicists today follows his experimental lead by inferring the actual nature of particles and interactions from the frequently counterintuitive results they find.

## The task

In the pictures (left), there is a target hidden by a black cloud. To figure out the shape of the target, we shot some beams into the cloud and recorded where the beams came out. Can you figure out the shape of each target? What assumptions have you made when deciding the shape?

In modern laboratories, experimenters use computers to analyse and visualise the data.

In terms of the experimental setup above, describe how you, personally, perceive the world. Note that you have more than one way to ‘observe’ your surroundings.

- If the target is the object that you observe, what is the source?
- How does the ‘detector’ work?
- What are the limitations to what you can perceive?
- How can (or do) you use technology to improve your perception of the world?